

Radwa Khattab

Senior AI Engineer | Agentic AI | Multi-Agent Systems | LLMs | AI Platforms | MLOps | Ex-Microsoft

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Professional Summary

Senior AI Engineer with 6+ years of experience building production AI systems, specializing in Agentic AI, multi-agent systems, and LLM applications. At Microsoft, developed production agentic capabilities for Microsoft Shopping using A2A, MCP, Semantic Kernel, tool/API calling, and structured contracts. More recently, led the end-to-end development of a production AI agent and supporting platform using Azure OpenAI, Python, FastAPI, and PostgreSQL, covering retrieval and data access, guardrails, PII protection, failure handling, and cloud deployment. Experienced across RAG, AI platform architecture, MLOps, Docker, CI/CD, and the production infrastructure required to take AI systems beyond prototypes.

Technical Skills

AGENTIC AI & ORCHESTRATION: Agentic AI · Multi-Agent Systems · Multi-Agent Platform · Multi-Agent Workflows · Multi-Agent Orchestration · AI Agents · LangGraph · LangChain · AutoGen · Semantic Kernel · MCP (Model Context Protocol) · A2A Communication · Tool Calling · Human-in-the-Loop Workflows · Approval Gates · Intent Detection

LLMs, RAG & RETRIEVAL: LLMs · SLMs · Generative AI · LLM Applications · RAG · Vector Search · PostgreSQL · pgvector · Pinecone · Embeddings · Prompt Engineering · Retrieval-Based AI · Grounded Generation · Structured Outputs

LLM EVALUATION, GOVERNANCE & LLMOps: Golden Dataset Evaluation · Deterministic Policy Evaluation · Evidence Validation · LLM-as-a-Judge · Safety Guards · PII Filtering · Auditability · LangSmith · LLMOps · Monitoring & Observability · OpenTelemetry · Application Insights

AI / MACHINE LEARNING: Machine Learning · Deep Learning · NLP · Computer Vision · Vision Transformers · Time-Series Modeling · Anomaly Detection · Model Fine-Tuning · PyTorch · TensorFlow · Keras · OpenCV · CUDA · Azure ML · Azure ML Pipelines · MLflow · DVC · ONNX · Unsloth · Model Training · Model Evaluation · ML Experimentation · Data Pipelines · Model Deployment

BACKEND, DATA & AI SERVICES: Python · FastAPI · Flask · REST APIs · Pydantic · Asyncio · PostgreSQL · SQL · NoSQL · JSONB · Node.js · JavaScript · TypeScript · React · .NET · JWT Authentication · Microservices

CLOUD, MLOps & INFRASTRUCTURE: Microsoft Azure · Azure AI Foundry · Azure OpenAI · Azure Container Apps · Azure Data Factory · Azure Databricks · Azure DevOps · AWS · Docker · GitHub Actions · CI/CD · Cloud-Native Architecture · Private Networking · Private DNS · Azure Key Vault · MLOps

WORKFLOW AUTOMATION: n8n · Policy-Based Workflow Automation · Structured Decisioning · Tool Workflows · Human Escalation

DATA & ANALYTICS: PySpark · Pandas · NumPy · SciPy · Power BI · Data Cleaning · EDA · Data Analysis

DEVOPS & DEVELOPMENT: GitHub · GitLab · Azure DevOps · GitHub Actions · CI/CD · Docker · Automated Testing · Unit Testing · Integration Testing · Deployment Automation · Linux · Bash · YAML

PROGRAMMING: Python · SQL · KQL · C++ · C# · JavaScript · TypeScript · R

AI-ASSISTED DEVELOPMENT: Claude · GitHub Copilot · AI-Assisted Code Generation · Code Review · Test Generation · Debugging · Documentation

Professional Technical Experience

Senior AI Platform Engineer | Proxify – Sweden, Remote, Contract

May'26 — Aug.'26

- Led the **end-to-end architecture and development** of an AI-powered performance intelligence platform spanning **AI services, backend architecture, data modeling, analytics, cloud infrastructure, CI/CD, and testing**.
- Designed and implemented a scalable production-grade **AI coaching agent** using **Azure OpenAI** and **FastAPI**, supporting **intent detection, contextual retrieval, structured data querying, multi-step workflows, personalized reasoning, and grounded response generation**.
- Built an **intent-based contextual retrieval layer** combining user profiles, wellbeing dimensions, habits, health indicators, methodology content, FAQs, and root-cause drivers to deliver **personalized responses without vector embeddings**.
- Redesigned the data layer from a deeply nested **MongoDB architecture spanning 55 collections** to a normalized **PostgreSQL schema with 20+ relational tables**, redesigning **data models, relationships, indexing, JSONB usage, and referential integrity**.
- Implemented **AI service safety, privacy, and reliability controls**, including **PII protection, GDPR compliance, role-based database access, controlled data retrieval, validation, logging, auditing, and graceful degradation**.
- Rebuilt the legacy Ruby analytics engine as a Python-based **calculation and analytics microservice** supporting **performance, wellbeing, health, productivity, engagement, and turnover-risk analytics**.
- Exposed AI coaching and analytics capabilities through **FastAPI microservices**, using **Pydantic, Docker, and Azure Container Apps**.
- Designed **Azure DevOps CI/CD pipelines** with **PR validation, automated pytest execution, Docker/ACR builds, environment-specific deployments, health checks, and failure diagnostics**, supported by comprehensive **unit and integration testing**.
- Modernized the platform from an over-engineered **Databricks architecture to a streamlined PostgreSQL-based cloud architecture**, reducing **cloud infrastructure costs by approximately 99%** while improving maintainability and operational simplicity.

Applied & Data Scientist (Agentic AI Engineer) | Microsoft – Egypt, Hybrid

May'24 — May'26

- Contributed to Microsoft's **Agentic Shopping Platform**, developing and integrating **multi-agent workflows** for shopping experiences using **A2A (Agent-to-Agent) communication**, **Semantic Kernel**, and **MCP (Model Context Protocol)**-based integrations. The platform was deployed to **production and integrated with Copilot's backend**.
- Developed **Semantic Kernel plugins** and agent-callable capabilities to connect AI agents with shopping and product services, including the **Other Buying Options** experience and related enterprise workflows.
- Integrated agent workflows with **MCP servers and gateway-based communication**, enabling structured access to external capabilities and services within **production multi-agent AI workflows**, with **Pydantic-based schema validation** for structured inputs and outputs.
- Designed and implemented a fine-tuned **Small Language Model (SLM)** pipeline using **Azure Machine Learning**, supporting **10 distinct scenarios with configurable inputs without code or pipeline modifications**, enabling reusable model fine-tuning and inference workflows while reducing reliance on GPT-based inference.
- Contributed to onboarding **10M+ product variants** through an **Azure ML product-page extraction and data processing pipeline**, supporting the hierarchical **Product → Variant → Offer** model and expanding catalog coverage across **Microsoft Shopping and Copilot**.
- Implemented a **statistical price validation and anomaly detection method** using historical product prices to flag anomalous price updates before storage, improving **data quality and reliability** across shopping experiences.
- Built a cross-team analytics platform using **Azure Data Factory** and **Power BI**, providing automated product insights, monitoring, and debugging capabilities for product managers and engineering teams.

AI Engineer | Ejada – Egypt, Hybrid / Saudi Arabia, Remote

Jan.'23 — Jan.'24

- Developed a **time-series Transformer model using PyTorch** to predict Saudi stock prices up to **three days ahead**, using historical OHLC data with feature engineering, normalization, and scaling.
- Built an automated **fine-tuning and evaluation pipeline** using market data from **Yahoo Finance** and internal Saudi-market data, evaluating predictions with **MAE and $\pm 1\%$ price-tolerance accuracy**.
- Built the MVP end-to-end with **Node.js, REST APIs, Docker, data visualization, and an on-premises GPU environment**, exposing the trained model through a prediction API and demonstrating results with a **Bank Al Rajhi** stakeholder.

Machine Learning Engineer | Silverkey Technologies – Egypt, Hybrid

Jan.'20 — Dec.'22

- Proposed and developed a **Vision Transformer + Temporal Convolutional Network (ViT-TCN)** for **video-based multi-emotion recognition**, owning dataset preparation, model development, training, experimentation, and evaluation using **PyTorch**.
- Collaborated with **Zewail City and ITIDA** on the research program, contributing to the resulting **Springer-published research paper**.
- Developed **computer vision and video analytics** solutions for billboard detection, advertising analysis, and **3D spatial reconstruction** using **YOLO, PyTorch, OpenCV, image segmentation, and Blender**.

Selected Portfolio Projects

Enterprise App Modernization Multi-Agent System | Synthetic Portfolio Use Case

- Built a LangGraph-based multi-agent system with parallel architecture and security analysis, RAG using PostgreSQL/pgvector, evidence-grounded findings, bounded revision, and checkpointed human-in-the-loop approval.
- Implemented deterministic evidence validation and approval gates, sanitized audit traces, LangSmith observability, and a golden-dataset evaluation workflow integrated into CI.

Publications

[ViTCN: Hybrid Vision Transformer with Temporal Convolution for Multi-Emotion Recognition](#) · Springer · 2024

Teaching Experience (Part-time)

TA — Nile University & Cairo University · Session Lead — Udacity · ML Instructor — AMIT Learning

Education

Master of Science in AI, Computer Engineering — Cairo University · Expected 2027

Bachelor of Science in Computer Engineering — Cairo University · 2021 · Excellent (85%)

Languages

• **English:** C1—Advanced (TOEFL)

• **Arabic:** Native